SANIA NEMADE

+1 201-744-8925 | saniapnemade@gmail.com | linkedin.com/in/sania-nemade | github.com/SaniaNemade

SUMMARY

I'm a Graduate student pursuing Masters in Engineering Management at Stevens Institute of Technology, with a B.E. in Computer Engineering and internship and project experience building retrieval systems and LLM applications. Seeking entry-level AI engineer roles.

EDUCATION

Stevens Institute of Technology | Hoboken, NJ — ***Expected May 2027***
Master of Engineering in Engineering Management — *GPA: 4.0/4.0*
St. Francis Institute of Technology | Mumbai, India — ***2021 - 2025***
Bachelor of Engineering in Computer Engineering — *GPA: 8/10*

SKILLS

Tools & Frameworks: Python, React, LangChain, LangGraph, LangSmith, Tool Calling, MCP
ML & Data: Scikit-learn, XGBoost, Pandas, NumPy, SQL, Matplotlib, Power BI, Tableau, Excel
Search & Retrieval: RAG, Hybrid Search, Sentence Transformers, FAISS
Backend & Infrastructure: FastAPI, Django REST, PostgreSQL, Docker, Git
Database: pgvector, ChromaDB, Qdrant, SQL

EXPERIENCE

Graduate Analyst - Systems Engineering | Stevens Institute of Technology — *May 2026 - Present*
- Pulled **1,000+** live job postings via API, stored them in SQL, and ran semantic search to match in-demand skills against **30+** Systems Engineering courses.
- Built a Tableau dashboard from the analysis that cut curriculum review meeting time from **3** hours to **1** hour.

AI Developer Intern | X eleva Group — *Jun 2026 – Aug 2026*
- Restructured a **10M+** row product dataset into a **3**-level category hierarchy and encoded product descriptions with sentence embeddings, reducing semantic search retrieval time by **30%**.
- Built a supplier-matching pipeline combining hard filters, pgvector HNSW similarity search, and keyword re-ranking to narrow **10M+** products to the top **10** supplier matches per buyer request.

ACADEMIC PROJECTS

Causal Studio - LLM for causal analysis | Python, OpenAI API, LangChain, DoWhy, Langsmith — *Summer 2026*
- Causal analysis in behavioral health is complex without statistical support; built an agentic dashboard that indexes **70+** research papers via RAG and uses GPT-4o to guide scientists through confounder selection, time windows, and matching methods using real-time wearable sensor data with **50k+** rows.
- Integrated DoWhy for causal estimation to trace every agent decision, turning days of manual literature review and study design into a single guided session.

QuackQuery | Python, Streamlit, ChromaDB, Sentence Transformers, Groq, PyPDF — *Spring 2026*
- Built a university-document QnA system that answers questions over PDF/DOCX files with source citations and supporting evidence passages.
- Combined **BM25** and semantic retrieval using **Reciprocal Rank Fusion** and benchmarked hybrid, semantic-only, and keyword-only retrieval on a **32-question eval set**, with hybrid performing best.

RoomieMatch | Python, Flask, FastAPI, Scikit-learn, MongoDB, React Native, Faker — *Winter 2025*
- Built a roommate-matching application that scored **1,000+ test profiles** based on lifestyle, location, and budget preferences.
- Developed a hybrid matching algorithm using **cosine similarity and K-Means clustering** to rank compatible roommates and integrated location-based filtering with Google Maps.

Automated Diagnosis of Autoimmune Diseases | Python, XGBoost, Scikit-learn, Pandas, SHAP, Matplotlib — *Fall 2025*
- Trained and compared **4** ML models to predict 6 autoimmune diseases from **12,500** patient records; XGBoost achieved **86.3%** accuracy and **0.88** macro F1.
- Used SHAP to identify the 15 most influential features and ran an ablation without antibody markers to test how well the model predicts from other clinical data alone.

Knowledge Platform | Python, Django REST, Gemini 2.5 Flash, Docker, PostgreSQL — *Spring 2025*
- Implemented a query-processing workflow that classifies incoming questions, routes them through hybrid vector/full-text retrieval, validates generated answers with a second LLM call, and caches results in Redis.
- Added query decomposition and caching to reduce redundant LLM calls by **40%** during load testing.